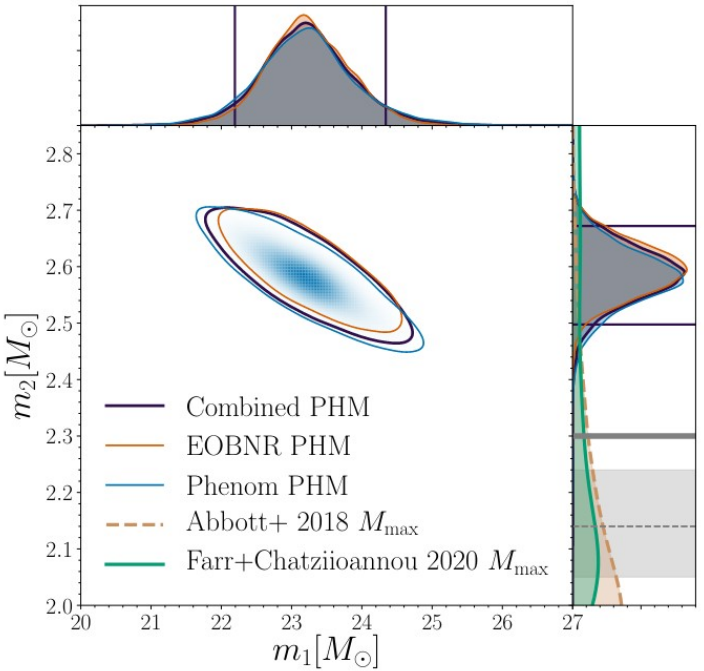
Highly Unequal Mass Black Hole Mergers In Active Galactic Nuclei

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In collaboration with: Evgeni Grishin | Alessandro Trani | Fiona Panther | Olga Pietrosanti

GW190814

- Gravitational wave event with most extreme mass ratio ever observed.
- Mass ratio q=m2/m1~0.1
- This event, and several other unequal mass BBH mergers are difficult to explain.



Credit: LIGO collaboration (2020)

Why most channels predict more equal mass

Isolated channel

Isolated binary:

- Binary black holes are formed from the collapse of massive stars in stellar binaries.
- If binary star masses are unequal, mass transfer phase from larger star (partially) equalises masses.
- Stellar evolution codes very predict mass ratios below q~0.25 should therefore be very rare.

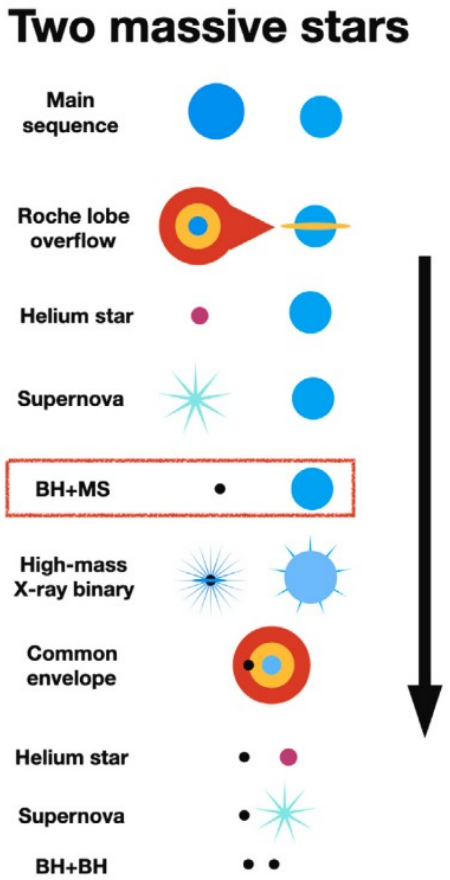


Image credit: Kareem El-Badry

Why most channels predict more equal mass

Dynamical channel

Globular cluster

- Dynamical formation of BBHs from close encounters of BHs in globular clusters.
- Binaries single interactions preferentially eject the lighter mass object.
- Even lower expected rate of GW190814 like events.

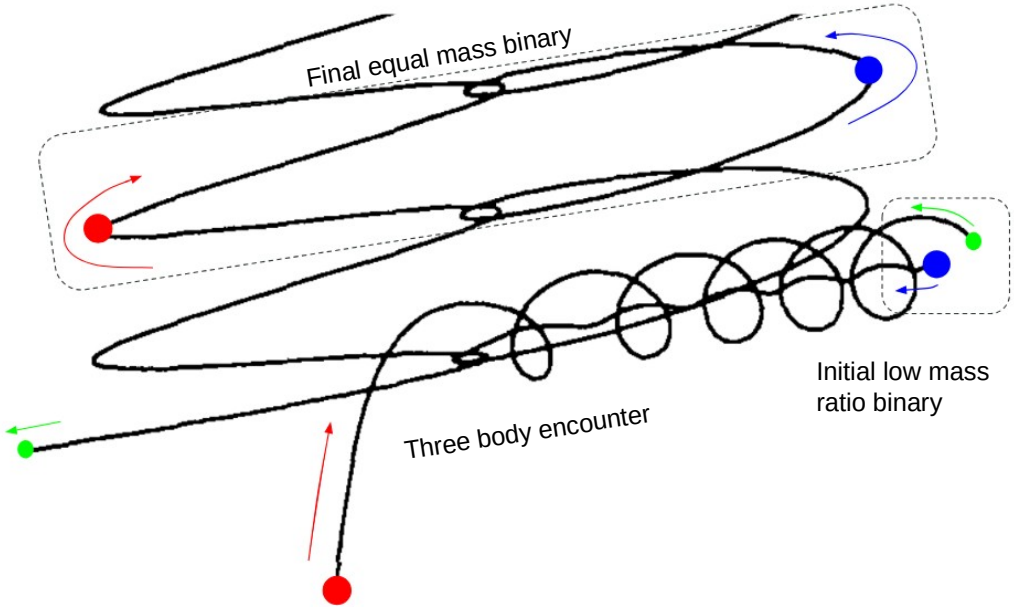


Figure adapted from Sigurdsson & Phinney 1993

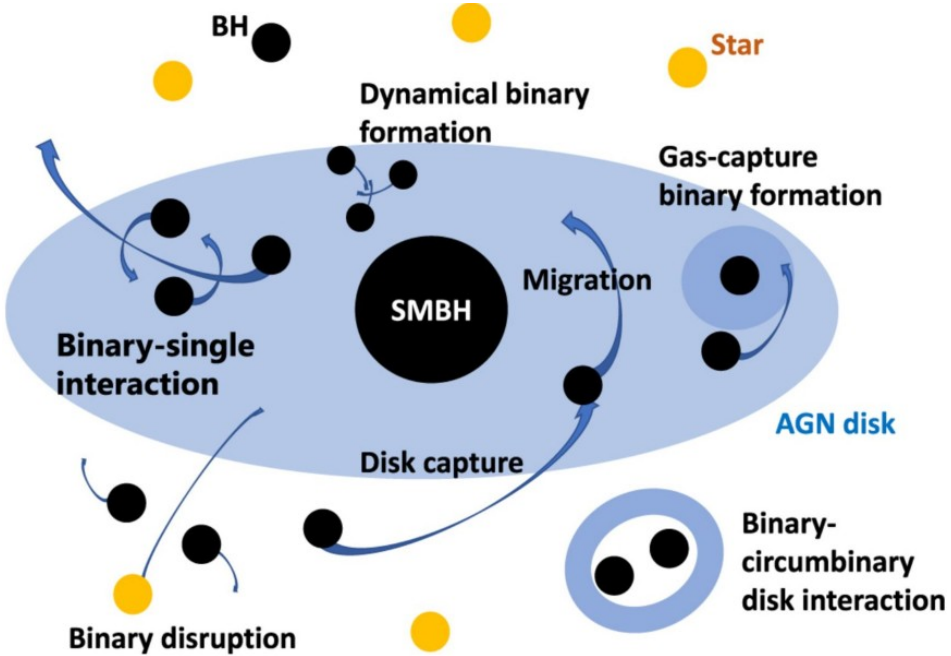


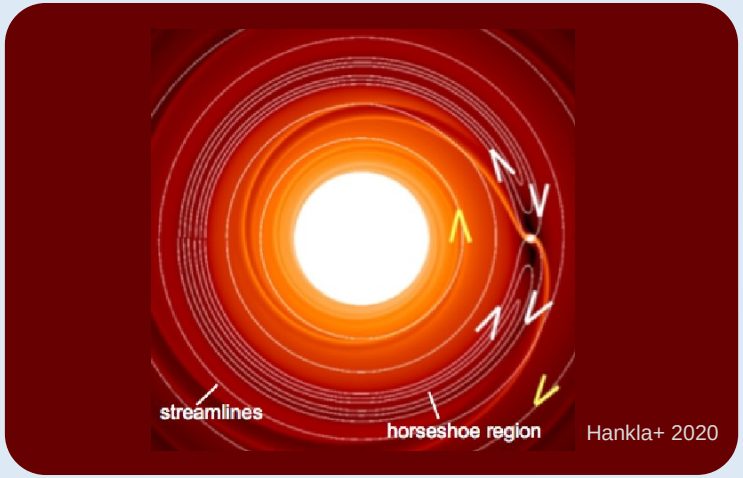
Image credit: Tagawa+ 2020

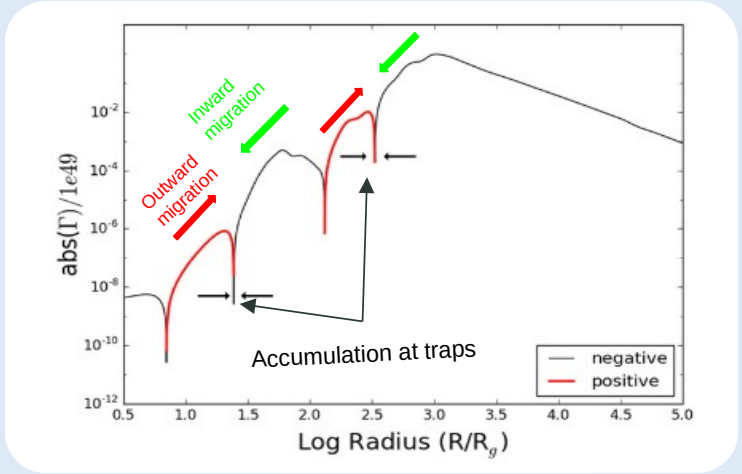
Hierarchical Mergers in AGN

- Hierarchical merging in Active Galactic Nuclei (AGN) accretion disks is a proposed solution.
- BHs in AGN sink toward center of disk, form binaries, and rapidly merge.
- Merged black hole could additionally merge with other BHs (hierarchical mergers).
- Proposed by many to explain GW190814, as well high mass events like GW190521 and GW231123.
- Gas rich environment could produce EM counterpart to GW emission.

Black holes in AGN disks migrate

- Massive object (black hole) perturbs AGN disk, generating spiral arms.
- Unequal gravitational attraction between BH and arms results in migration.
- At locations where the dominating arm swaps, migration stops (migration trap).
- Build up of black holes, facilitating many interactions and mergers.





Adapted from Bellovary+ 2016

Thermal torques

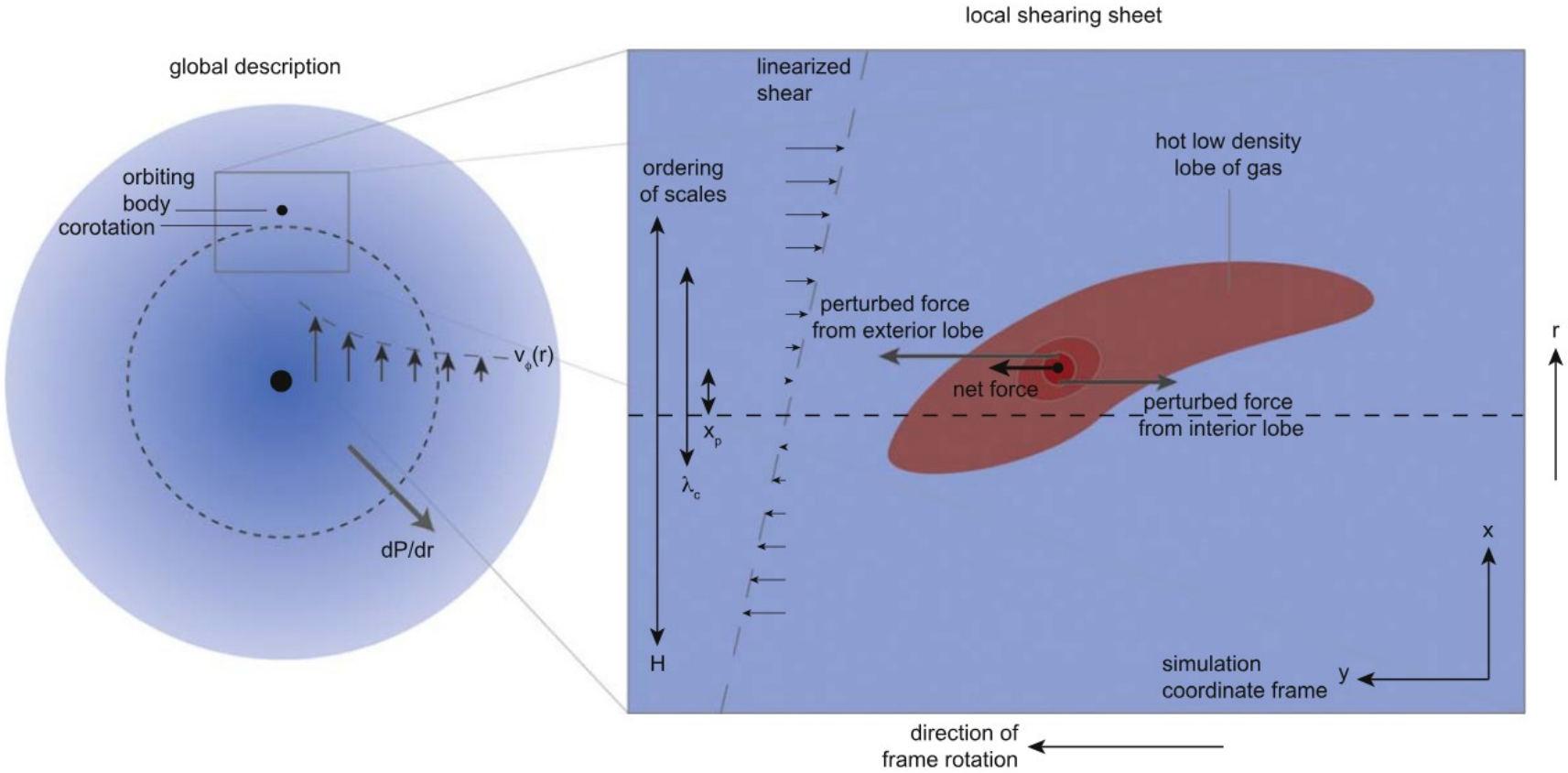


Image credit: Hankla+ 2020

Large BHs open gaps in the disk

- Large BH can form gaps in the disk, preventing it from becoming trapped.
- Mass dependent trap locations could prevent highly unequal mass mergers like GW190814.

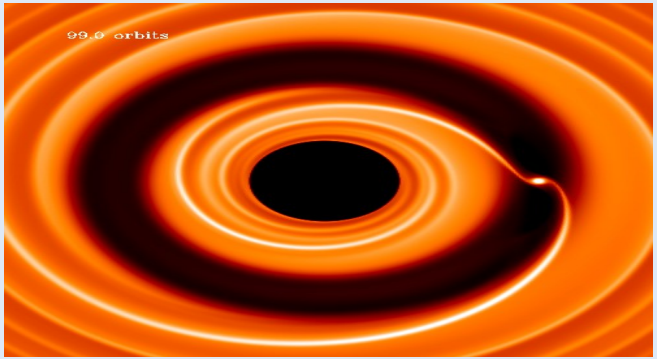
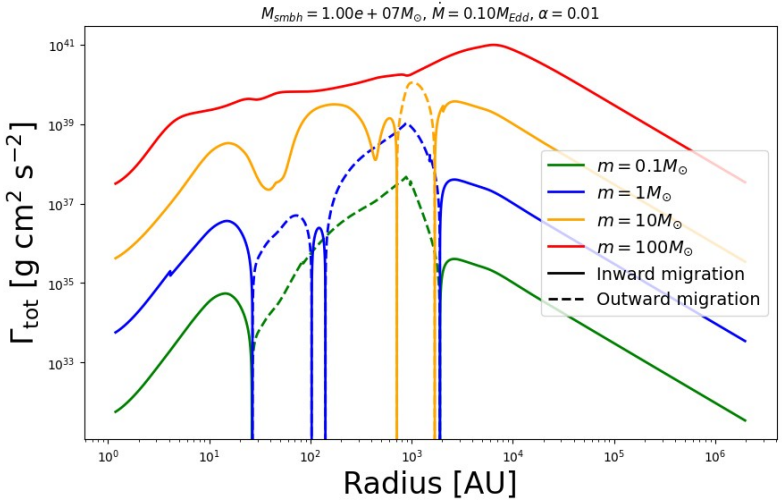


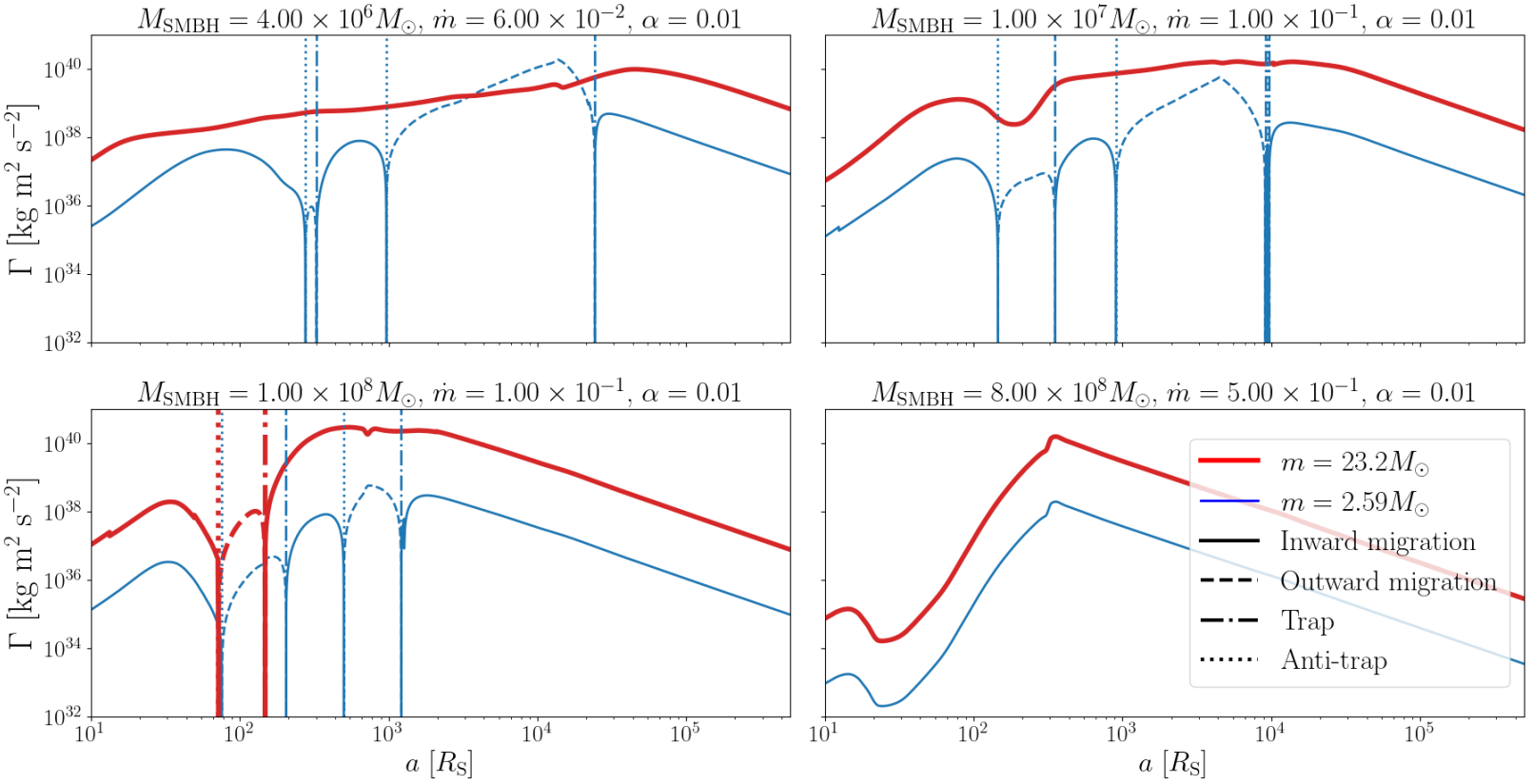
Image credit: Frederic S. Masset

Torque suppression from gap opening

$$\Gamma_{tot} = \Gamma_i + e^{-K/K_t}\Gamma'_{th}$$



Torque profiles

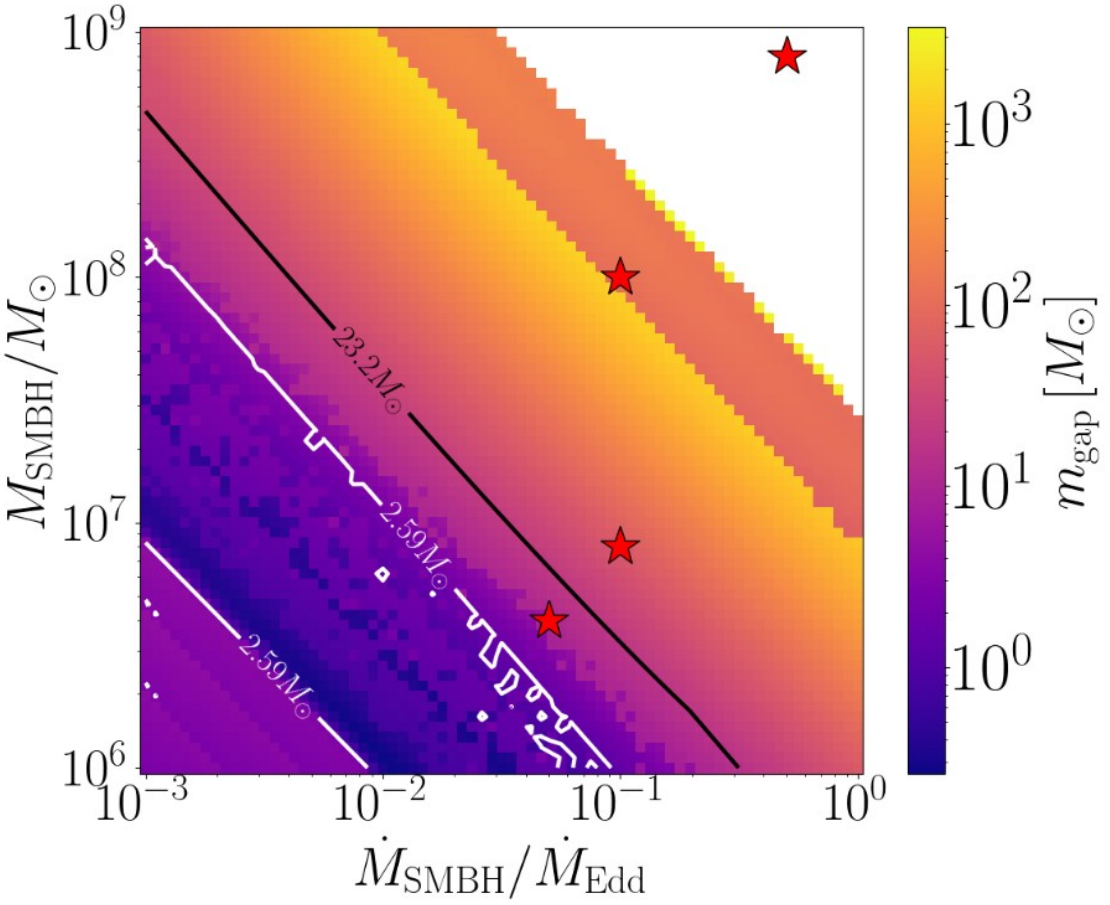


AGN disc parameter space

$$L_{\rm AGN} \propto M_{\rm SMBH} \cdot \dot{M}_{\rm SMBH}$$

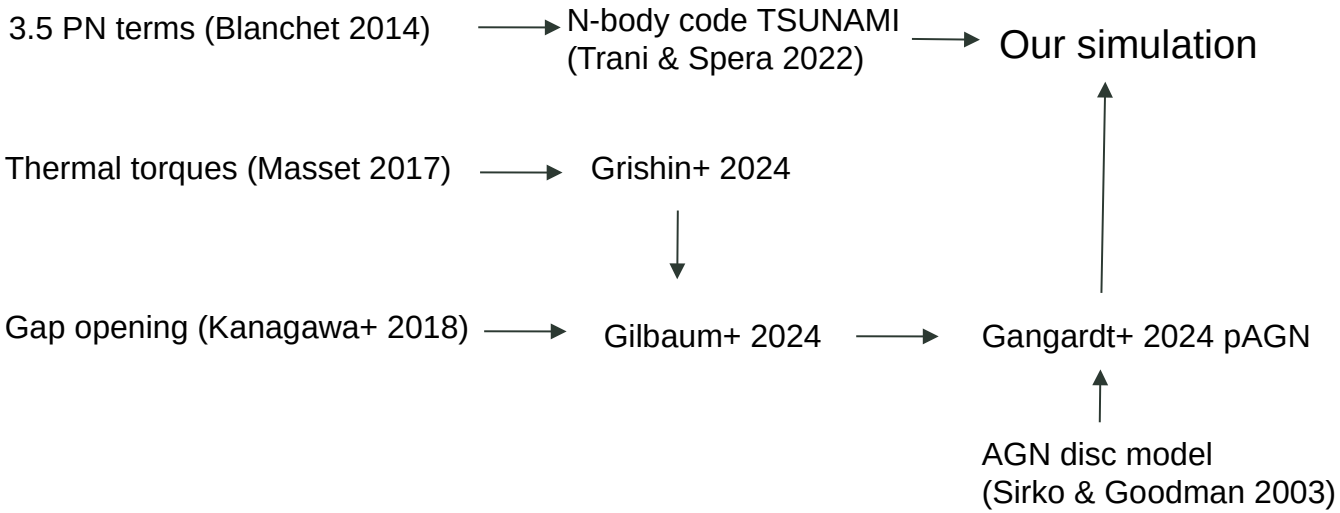
$\Rightarrow m_{\rm gap}$ is well parameterized by AGN luminosity

How does this affect merger rates for different AGN discs?

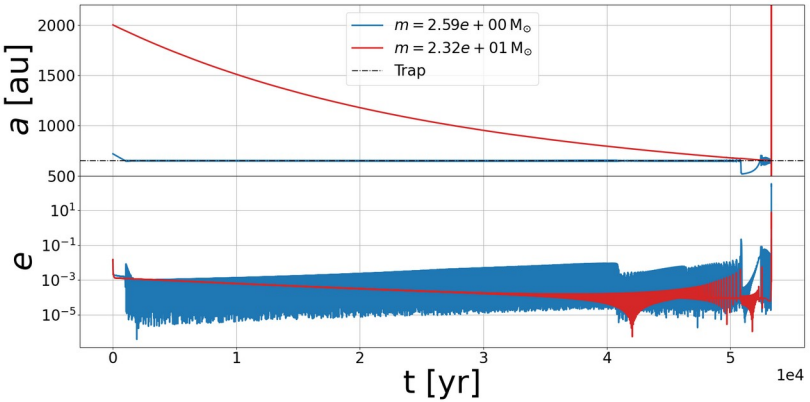


Code setup

- Our code couples gas dynamics with N-body forces to capture BH dynamics over long and short time scales.
- N-body code includes relativistic corrections up to 3.5PN.
- Uses semi-analytic prescriptions for migration torques that accurately fits results from 3D hydro sims.
- With AGN disk model, allows for accurate study of sBH dynamics across the AGN disc parameter space.

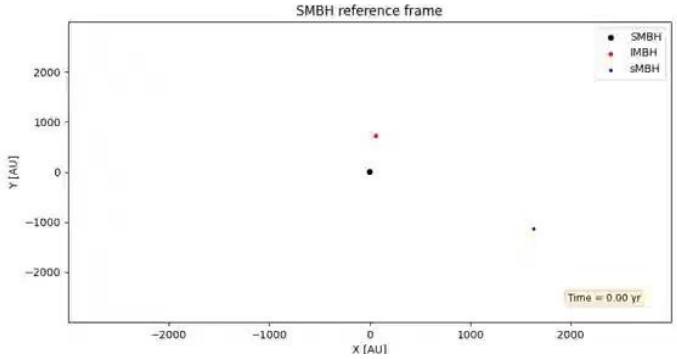


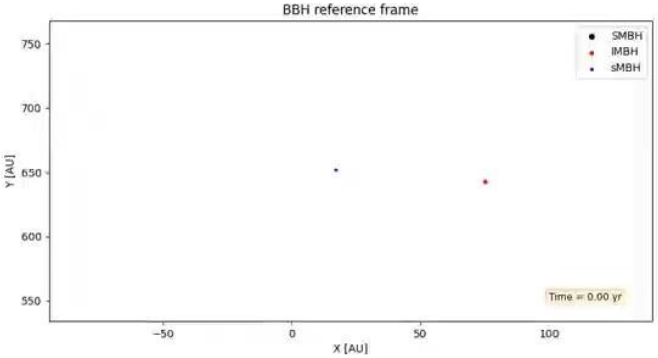
Mergers at traps in our simulations



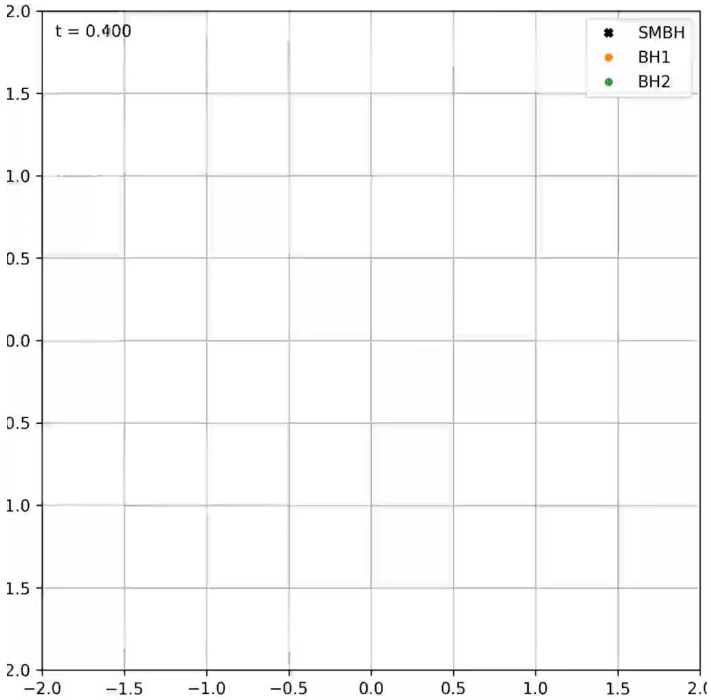
Relativistic N-body code (TSUNAMI)

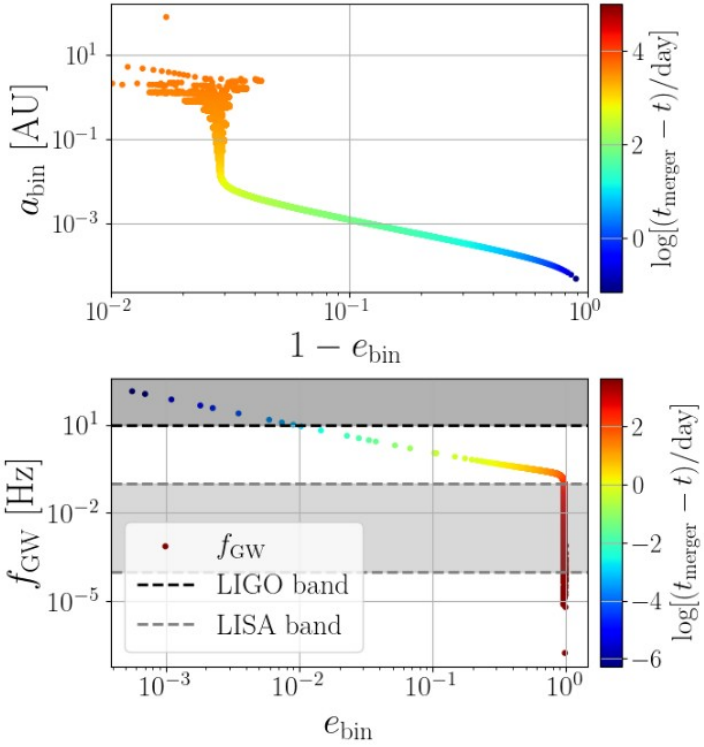
AGN disc + gas torque model (pAGN)



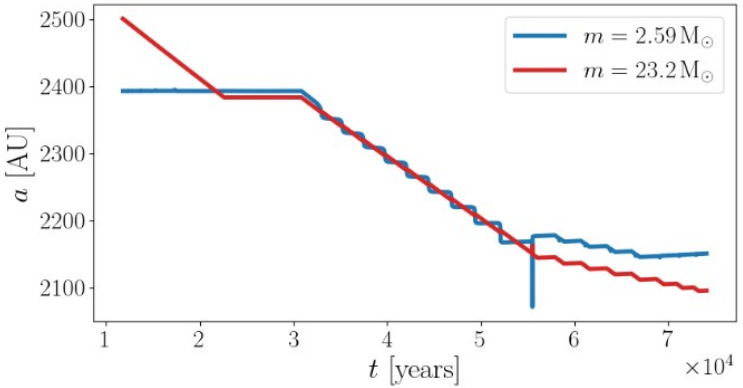


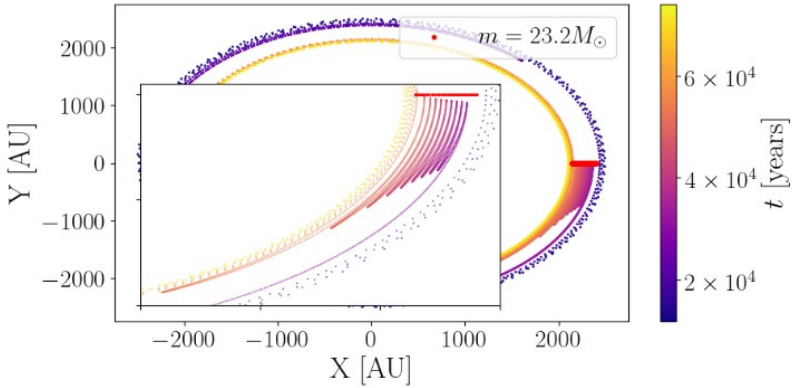
Eccentric mergers

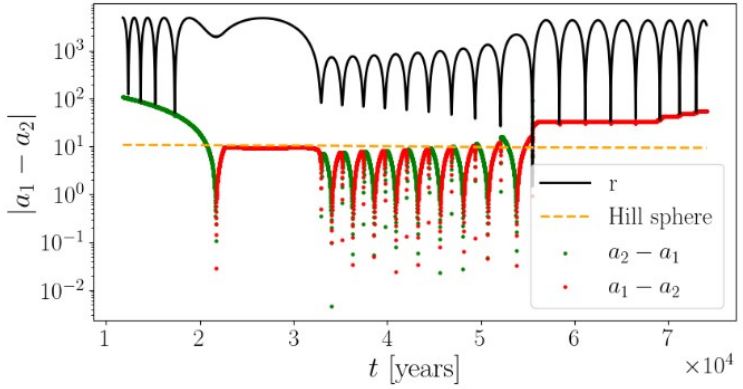


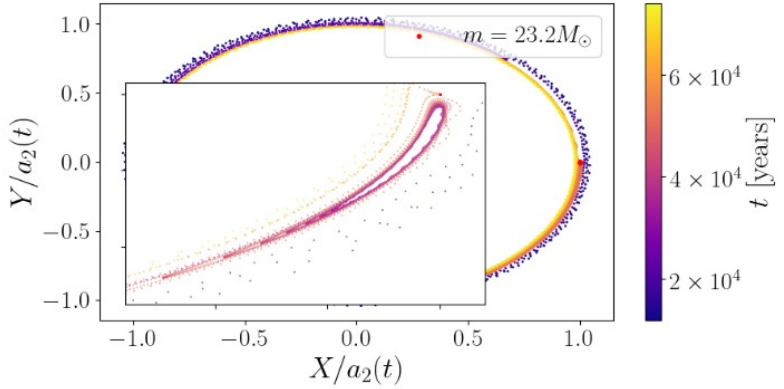


Tadpole orbits

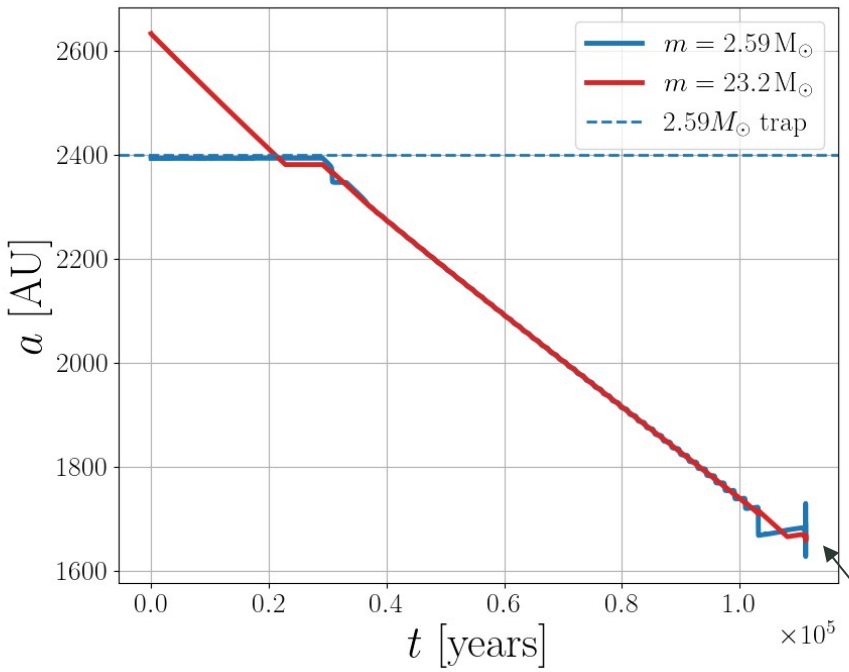




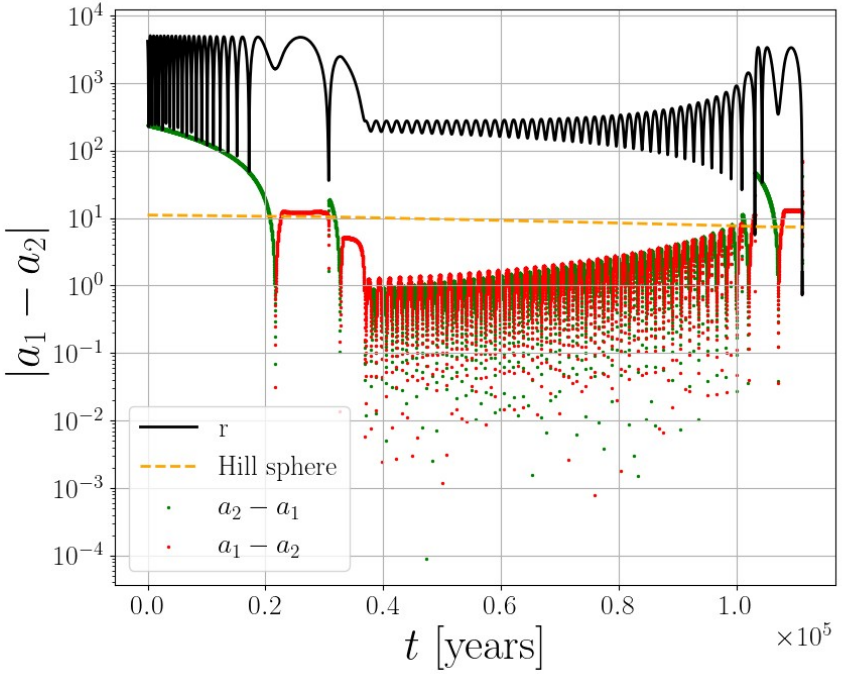




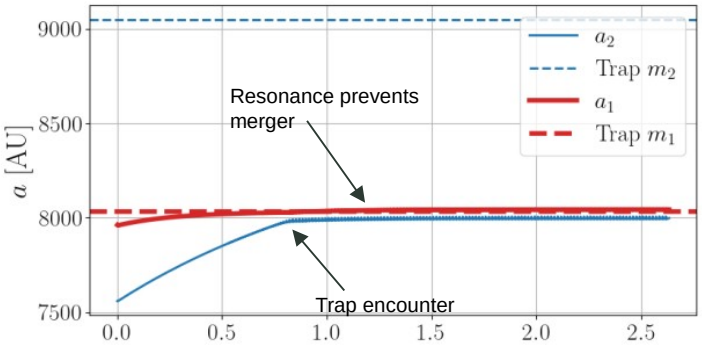
Tadpole breaking

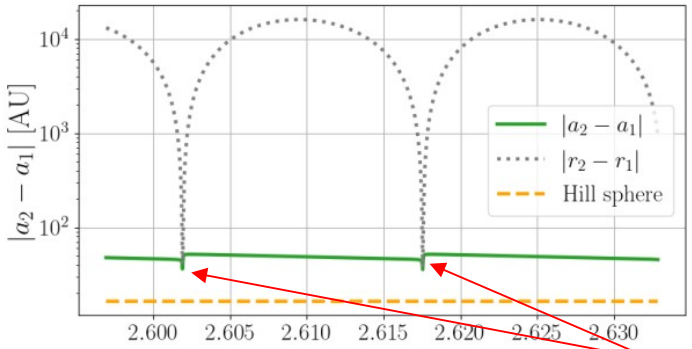


Merger

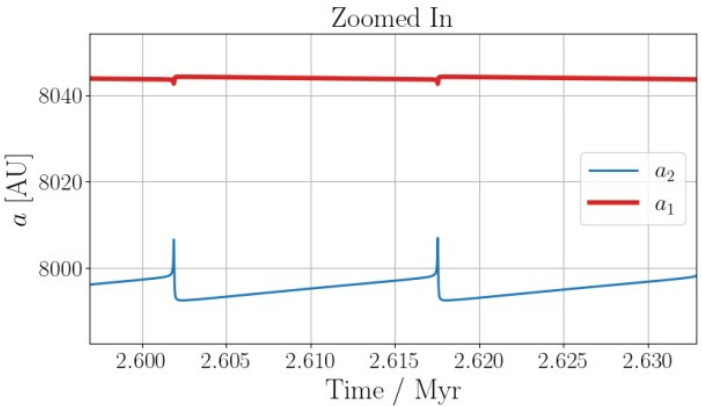


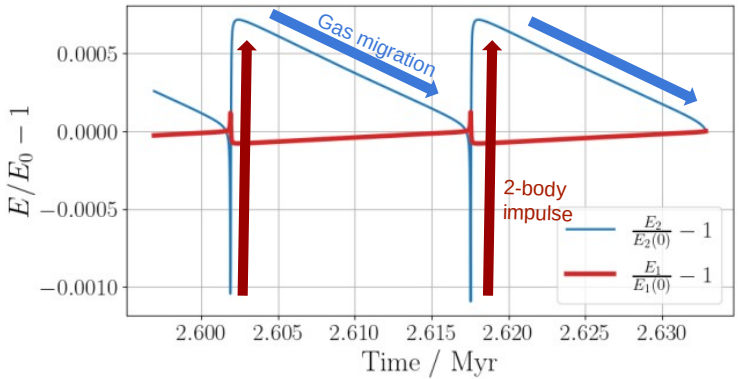
Resonance blocking



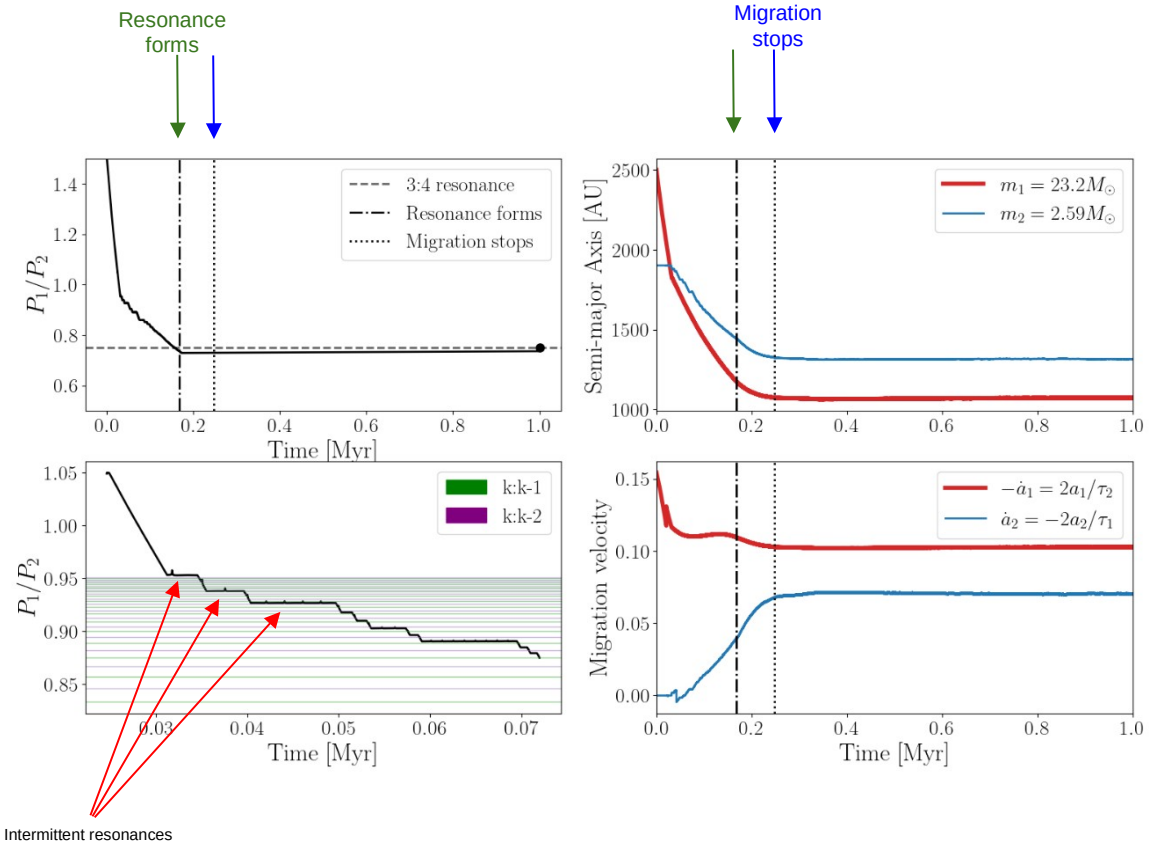


Close encounters



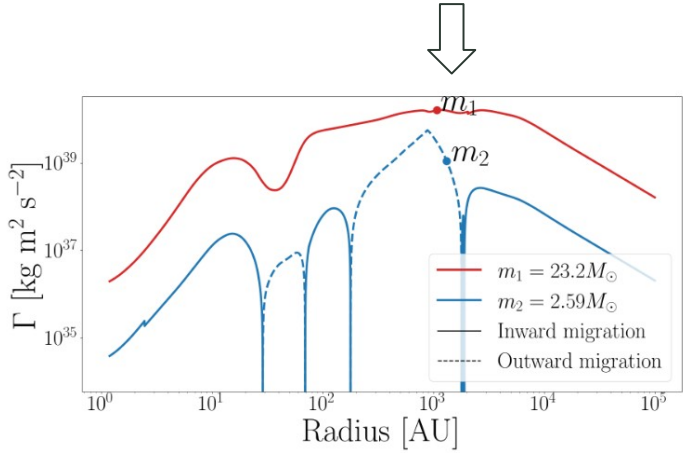


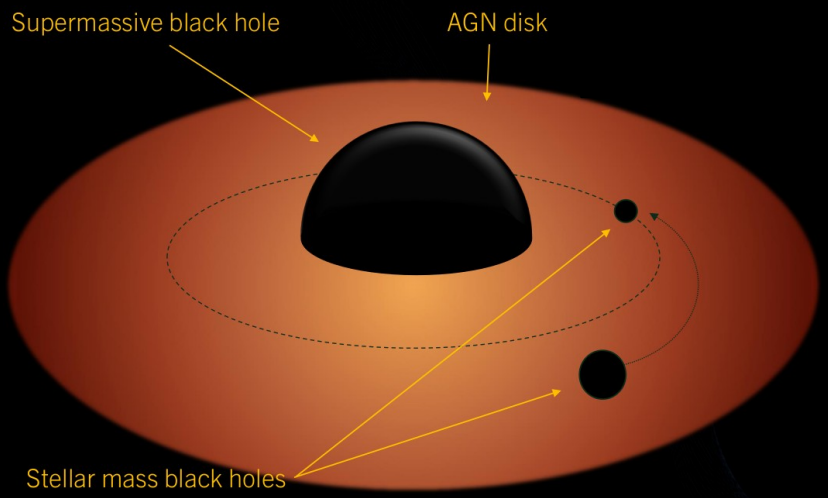
“Resonance trap”



Intermittent resonances

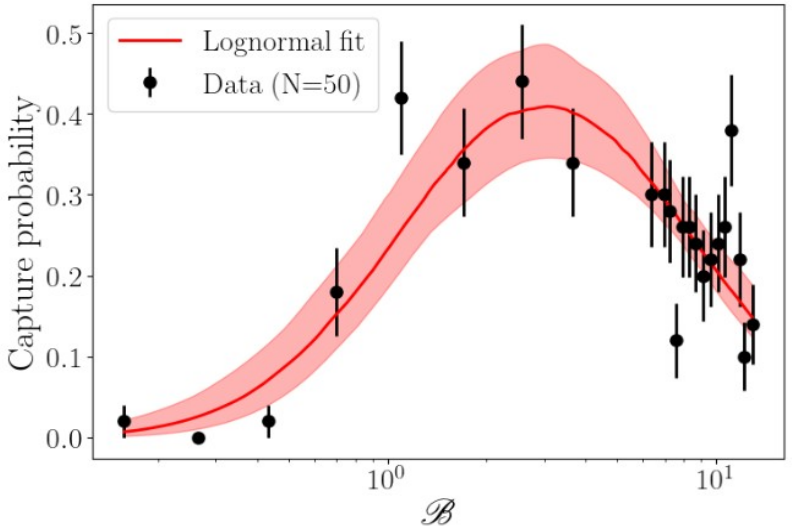
Final positions

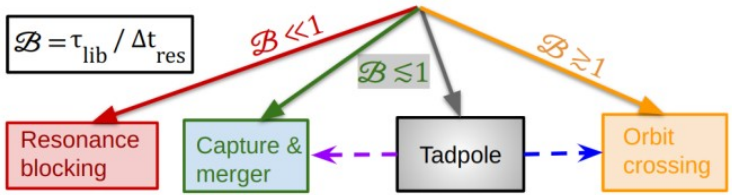




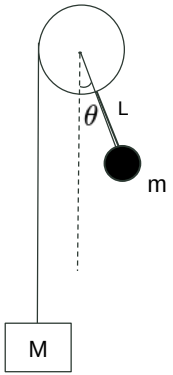
=> Limits on the AGN capable of producing high mass ratio events like GW190814

What determines the encounter outcome?





Resonance capture



$$\mathcal{H}_{\rm pend} = \frac{p^2}{2(m+M)} - mgL\cos\theta - MgL\theta$$

$$M \gg m \implies \text{circulation}$$
$$m \gtrsim M \implies \text{libration}$$

$$\text{Oscillation period } \tau_{\rm lib} = \sqrt{g/L}$$

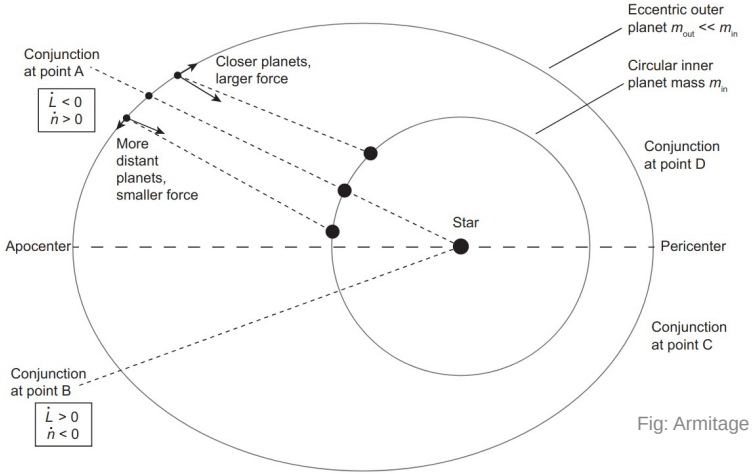
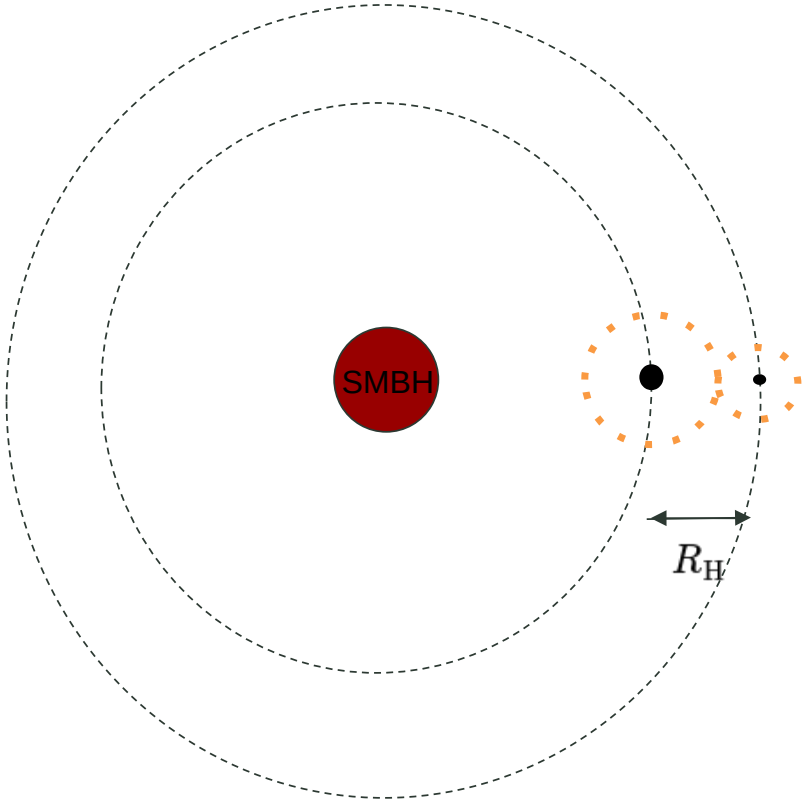


Fig: Armitage 2020

$$\mathcal{H}_{\rm res} = -\tilde{\Psi}^2 - 2\sqrt{2\tilde{\Psi}}\cos\psi + 3(\delta+1)\tilde{\Psi}$$ Batygin (2015)

$$\mathscr{B} \equiv \frac{\tau_{\rm lib}}{\Delta t_{\rm res}} \approx \frac{P_2}{\tilde{\tau}_{\rm mig}} \left(\frac{M_{\rm SMBH}}{m_1+m_2}\right)^{4/3} \frac{1}{4k^{2/9}(\sqrt{3}|f_{\rm res}^{(1)}|)^{4/3}} \ll 1$$



Resonance capture

- Guarantees capture for large k, i.e. when $\frac{P_2}{P_1} = \frac{k}{k+1} \to 1$
- To prevent resonance overlap and onset of chaos, set

$$k_{\max} = \frac{2}{3}\left(\frac{3M_{\rm SMBH}}{m_1 + m_2}\right)^{1/3}$$

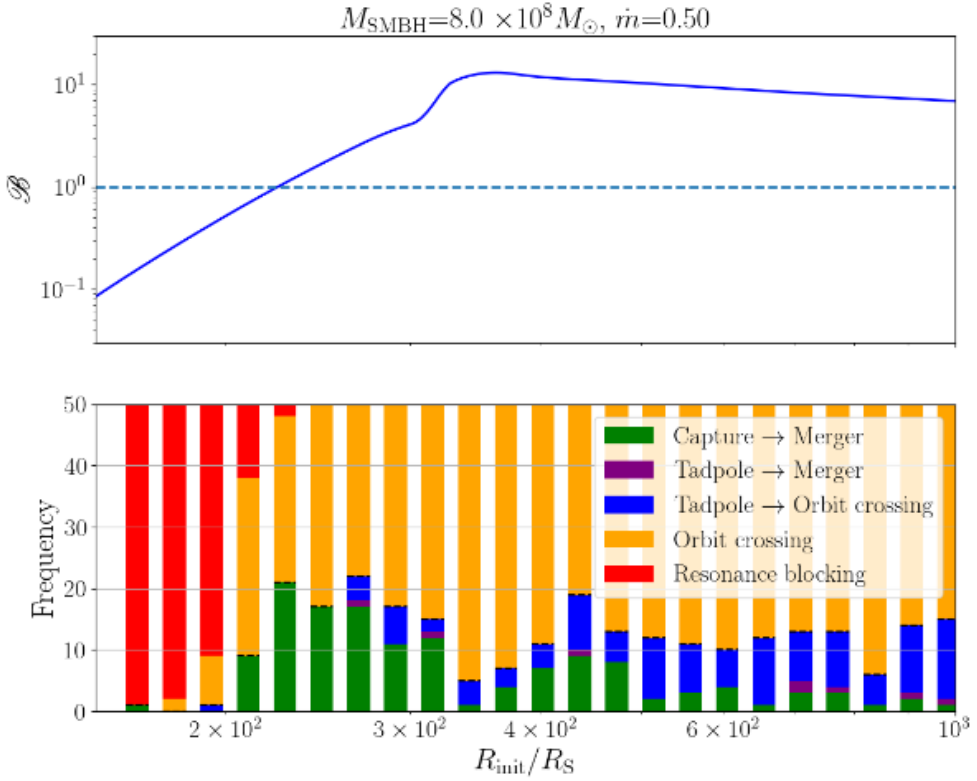
- This gives an upper bound, such that capture into some resonance is guaranteed (not probabilistic).

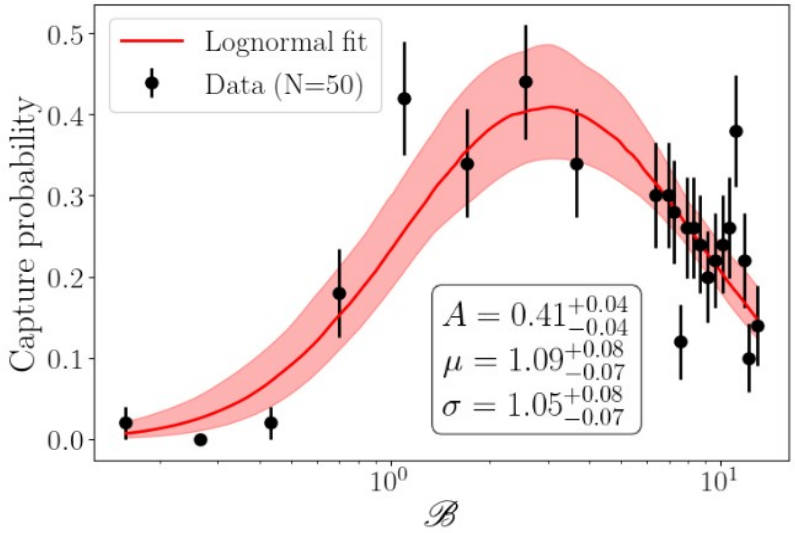
$$k = k_{\max} \implies \mathscr{B}^{-1} \sim \frac{\#\text{Orbital encounters}}{\text{Hill sphere crossing}}$$

- Hence orbit crossing (without captures) are guaranteed in the opposite limit to resonance capture.
- Binary capture, and subsequent merger, can thus only occur when

$$\mathscr{B} \sim 1$$

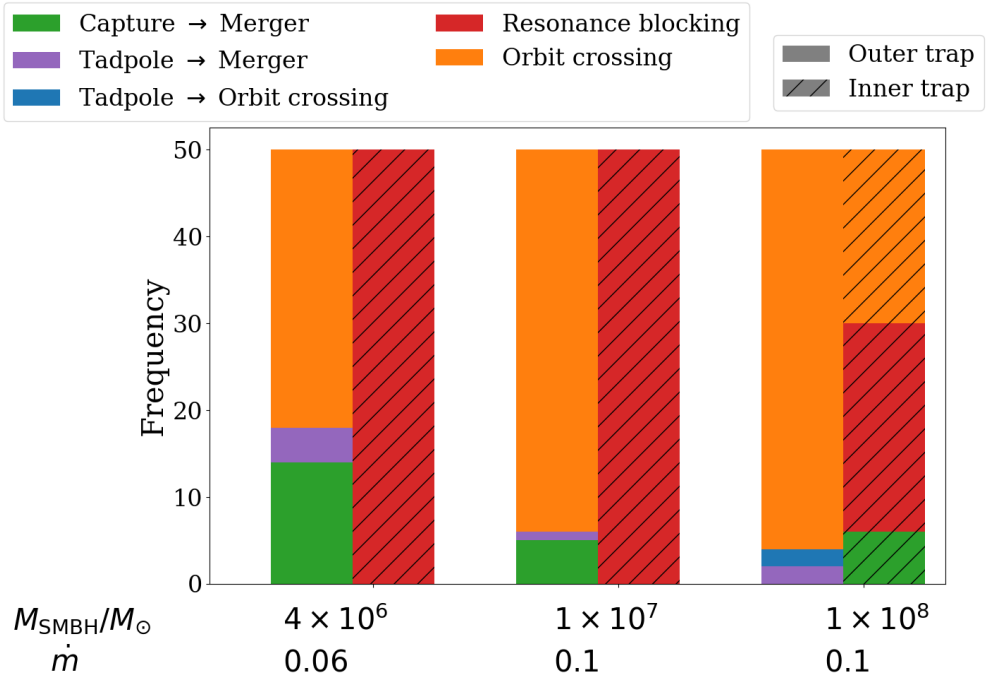
Check with simulations

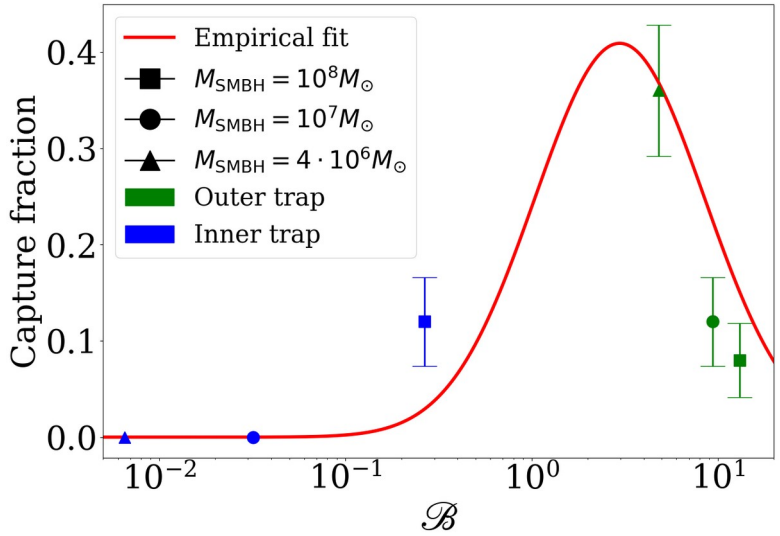




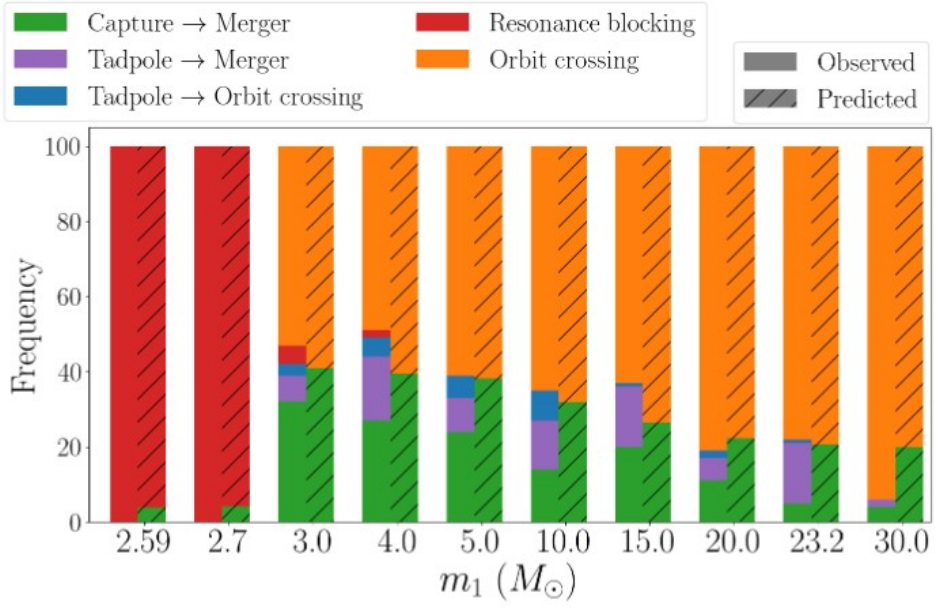
$$P(\text{capture}|\mathscr{B}) = \text{A}\exp\left(-\frac{(\ln\mathscr{B}-\mu)^2}{2\sigma^2}\right)$$

Varying disc parameters

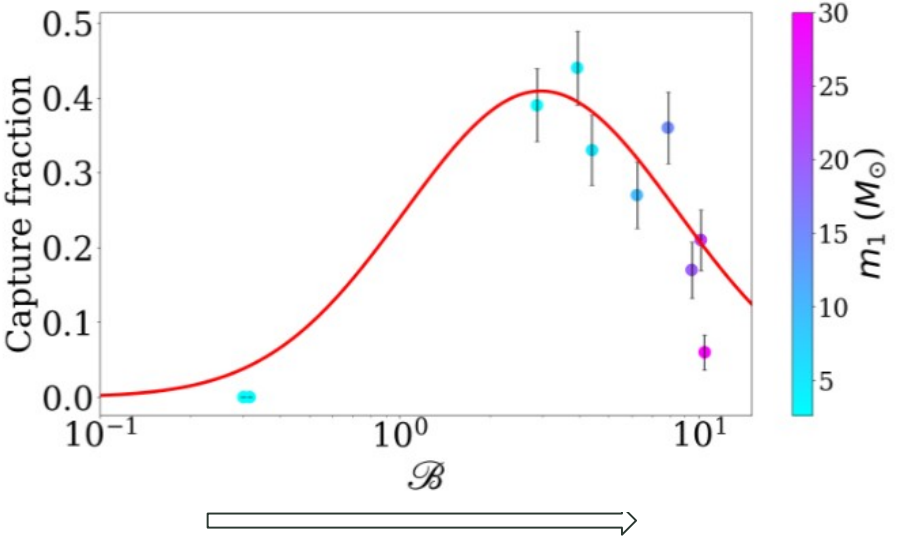




Varying mass ratio

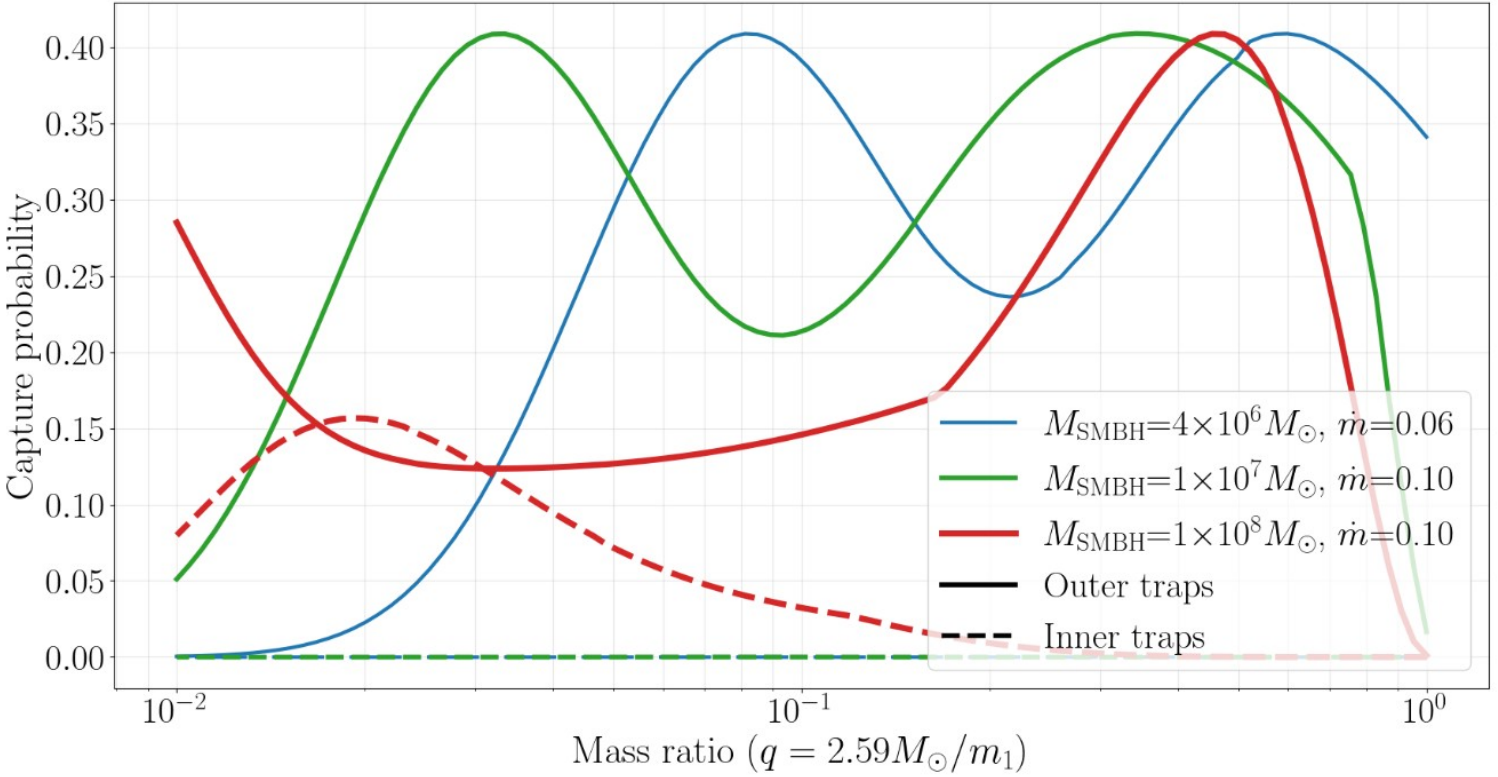


$m_2 = 2.59 M_\odot$ fixed

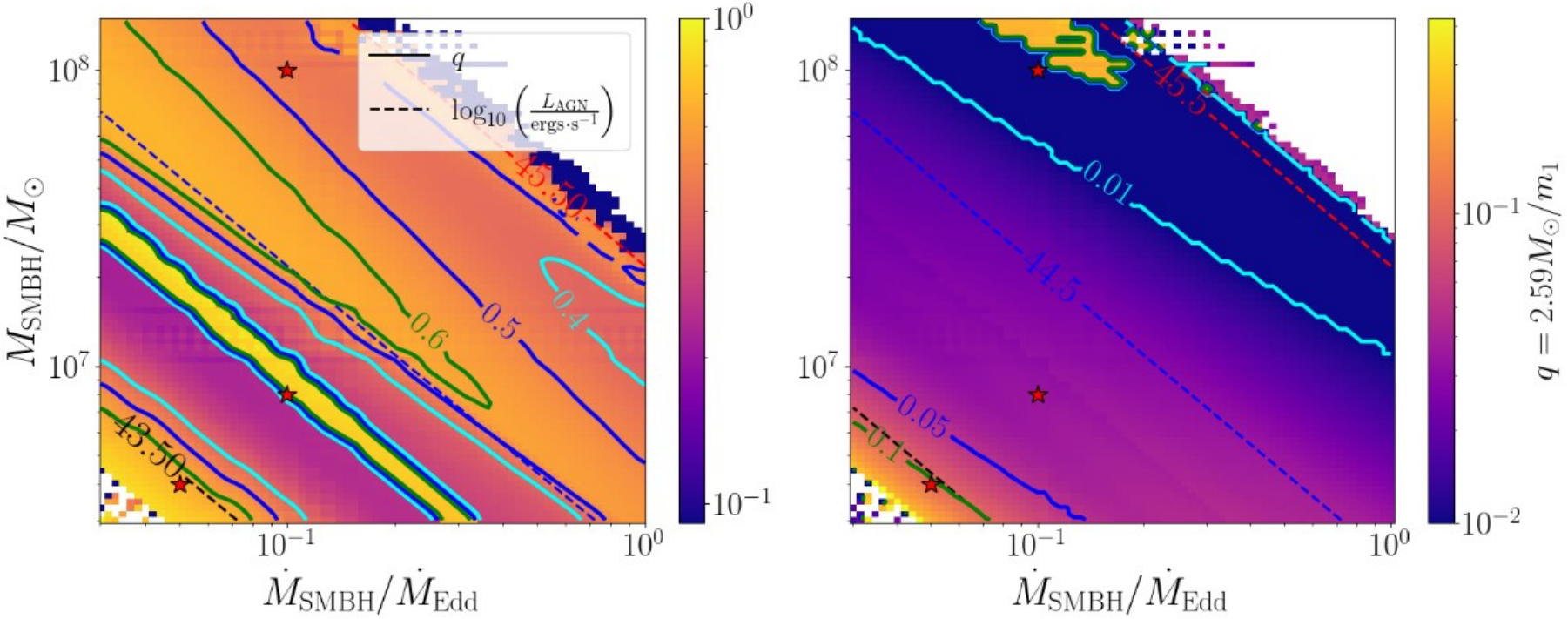


Increasing m1

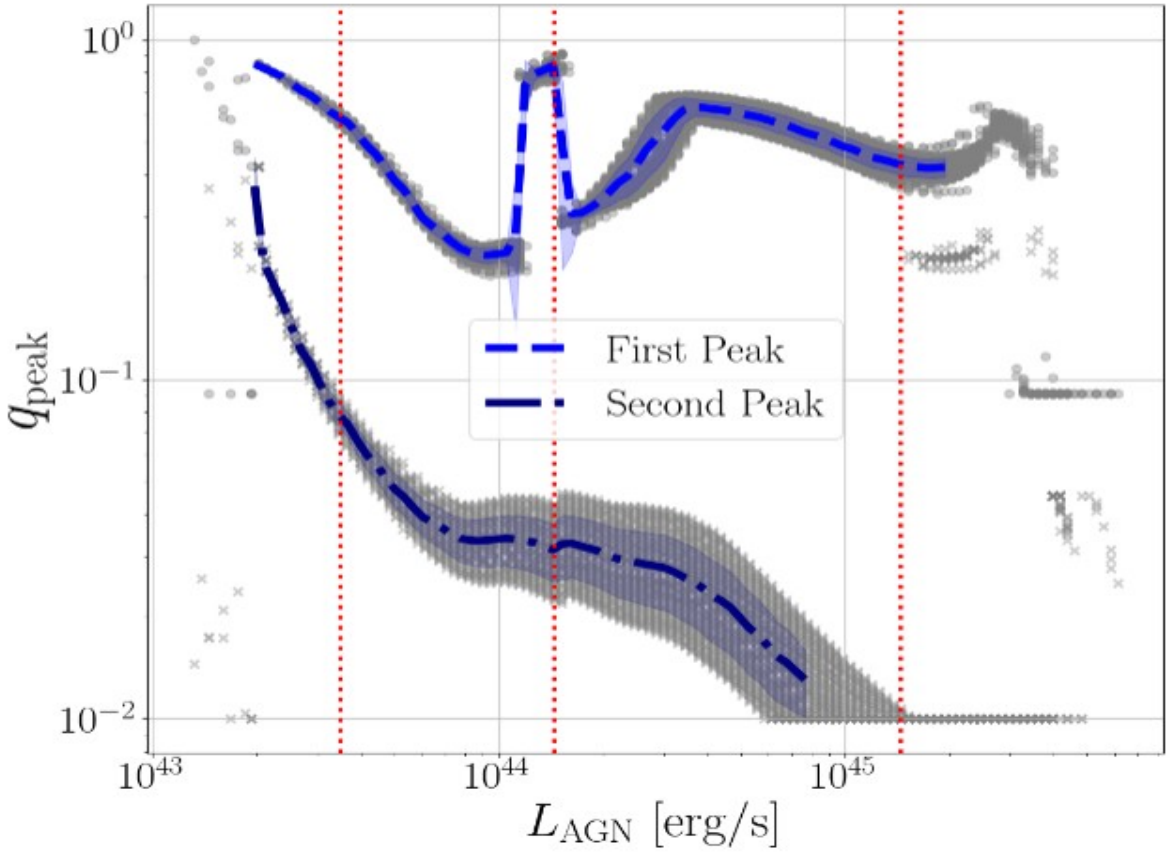
Trap capture probability disc dependence



Optimal q across AGN parameter space



Correlation of optimal q with AGN luminosity



Summary and future prospects

arXiv:2511.09129

- Coupled relativistic N-body code with AGN disc models to produce realistic simulations of highly unequal mass black hole encounters.
- Despite the complicated dynamics ignored in previous studies, we identified that binary formation probability scales simply with a parameter which depends on:
  - The black hole masses,
  - AGN disk parameters,
  - Encounter location.
- Gives insight into which AGN galaxies could efficiently produce events like GW190814 mergers.
- Refinements could lead to a novel approach of ranking AGN galaxies based on host galaxies, not possible with other channels without an EM signal.
- Potential implications for cosmology and other Physics tests with gravitational waves